

Design of Vedic Mathematics based 16 bit MAC unit for Power and Delay Optimization

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Abstract—Multiply and Accumulate unit is the essential component of any Digital signal processor. In designing such a unit, it is significant to consider various design parameters such as power, area, and delay for a given application. Also, an efficient multiplier and adder units play a vital role in deciding the performance of MAC unit and hence DSP. In this paper, the design of 16 bit MAC unit using 16 bit Vedic multiplier and a 32 bit spanning tree adder is presented. The Vedic multiplier, in turn, is designed utilizing carry save and spanning tree adders by applying the concept of Urdhva Tiryakbhyam sutra. This entire design is coded in Verilog HDL using the tool Xilinx Vivado 2018.3 simulator with a targeting Zedboard (xc7z020-3clg484) for synthesis and implementation. This proposed multiplier design is compared with design of the existing 16 bit Vedic multiplier which is made up of Carry Save Adders. The results showed that there is an improvement in delay and power by 30% and 21.1% respectively compared with the existing design.

Keywords—Vedic mathematics, high speed, spanning tree adder, MAC unit.

I. INTRODUCTION

The fundamental block used in various complex signal processing applications is Multiply and Accumulate (MAC) unit. In the computational applications such as Discrete Fourier Transformation (DFT), Fast Fourier Transformation (FFT), Filtering, Convolution, Finite Impulse Response filters, etc. this unit plays a critical role in improving the efficiency and performance which makes the Digital Signal Processor (DSP) highly dependent on the MAC unit. As multiplier and adder are main parts of a MAC unit, they must be chosen wisely in order to achieve optimization both in power and delay.

Vedic mathematics is a practice retrieved from ancient Vedas of Hindu scriptures. According to Indian Monk Sri Bharati Krishna Tirthaji's research published in 1965, there are 16 Vedic mathematics sutras. These techniques reduce the complexity by cutting out excess steps while performing the operations. Hence, Vedic mathematics is one of the most efficient and refined mathematical systems. Among 16 sutras, Urdhva Tiryakbhyam (UT) sutra is widely used because of its faster calculative multiplication abilities. The Sanskrit word Urdhva Tiryakbhyam meaning vertically and crosswise and is preferred for smaller order numbers. This process of vertical and crosswise multiplication makes it easier to generate the partial product outputs. The traditional multiplication

procedures generate outcomes consuming certain delay which is not suitable for current trends. To reduce the delay and improve effectiveness of the signal processing calculations, Vedic mathematics based multiplication is preferred.

The UT sutra is used in this design as it supports pipelined computing with lesser delay by making it a faster multiplier. In order to achieve low delay functioning, Spanning tree adder is used which is a modified version of the Carry look ahead adder [1]. For the addition of more than two partial products at a time with less power and less delay, Carry Save Adder (CSA) is used.

In this paper, 16 bit MAC unit built using Vedic multiplier is designed which uses CSA and spanning tree adder along with lower order multipliers. This design is then compared with an existing system that uses the same multiplier logic but comprising CSAs. The paper consists of related works described in Section II, further in Section III, the design of MAC unit is elaborated. Results are discussed in Section IV and conclusion is made in Section V.

II. RELATED WORKS

Various researchers used CSAs and carry select adders in the design of Vedic multiplier for MAC unit and disclosed development in either delay, power or area [2], [3], [4], [5]. The Vedic multiplier (UT sutra) comprising Kogge - stone adder is realized [6]. The system showed improvements in terms of speed as the adder generated a carry signal in $O(\log N)$ time. Nikhilam Sutra is analyzed in terms of total calculation time and multiplication for inputs of different bit lengths [7]. The design of a Vedic multiplier, which exploited 4:2 compressors instead of standard full adders showed improvements in performance [8]. The MAC unit is built using modified Booth's algorithm and when related with the conservative algorithm resulted improvements in power and delay [9]. A 64 bit Vedic multiplier utilizing UT sutra and Han Carlson adder is related with conventional Booth and Array multiplier [10]. Pipelining technique is employed on Vedic multiplier (UT sutra) which resulted in usage of less power compared to non pipelined architectures [11]. A 32 bit multiplier is designed using booth and pipelined Wallace tree architectures along with the usage of carry select adder to reduce the delay [12]. The design of Wallace tree multiplier using 4:2 compressors and parallel prefix adders to reduce the number of stages to get the output for higher order multiplication is presented in [13]. From the research done,

Vedic multiplier is used because of its efficient computational structural design.

III. DESIGN OF VEDIC MATHEMATICS BASED 16 BIT MAC UNIT USING SPANNING TREE ADDER

Fig. 1 shows the design of existing 16 bit Vedic multiplier which uses two (02) 16 bit CSAs for adding the partial products. Also, four (04) 8 bit Vedic multipliers, each built from 4 bit Vedic multipliers and 4 bit CSAs are designed. The second CSA which can add three inputs does computation only on two inputs i.e., partial product from the most significant 8 bit vedic multiplier and higher order bits of the sum from first 16 bit CSA. Due to this reason, hardware elements in the second CSA become redundant. These redundant elements increase the delay, power and area of the entire design. So, in the proposed 16 bit MAC unit shown in Fig. 2, multiplier uses a 16 bit spanning tree adder in place of the second 16 bit CSA as shown in Fig. 3.

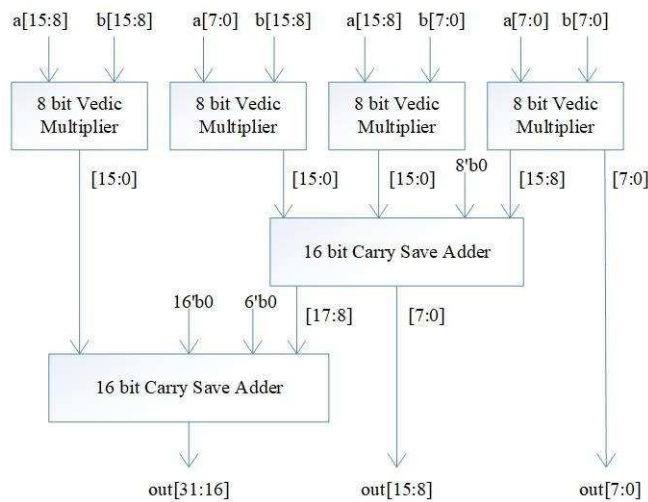


Fig. 1. Existing 16 bit Vedic multiplier using 16 bit CSAs.

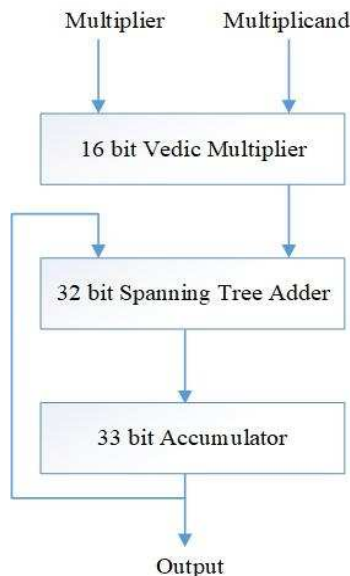


Fig. 2. Proposed 16 bit MAC unit using 16 bit Vedic multiplier and 32 bit spanning tree adder.

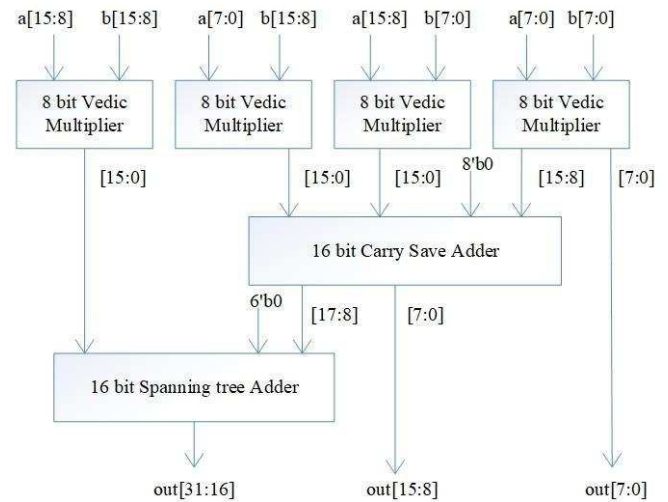


Fig. 3. Proposed 16 bit Vedic multiplier using 16 bit CSA and 16 bit spanning tree adder.

Both 16 bit CSA and 16 bit spanning tree adder are used to add the partial products generated from 8 bit Vedic multipliers. Similarly, the design of 8 bit Vedic multiplier is carried out. Whereas the design of 4 bit Vedic multiplier includes 4 bit CSA, 4 bit ripple carry adder and 2 bit Vedic multipliers as shown in Fig. 4. If 4 bit spanning tree adder is used as the second adder, only one carry will be produced by the propagate and generate bits and there will be no significance for this carry produced as it is not used again in the computation of product. So the hardware used for generating carry propagate and carry generate bits is of no use, hence a 4 bit carry select adder can be used instead of 4 bit spanning tree adder. In the 4 bit Vedic multiplier, the carry given as input to the second adder is always zero and so this carry select adder acts as a ripple carry adder by removing the additional hardware elements which are used when the input carry is one. The carry given as input to all the adders in all the multipliers is 0.

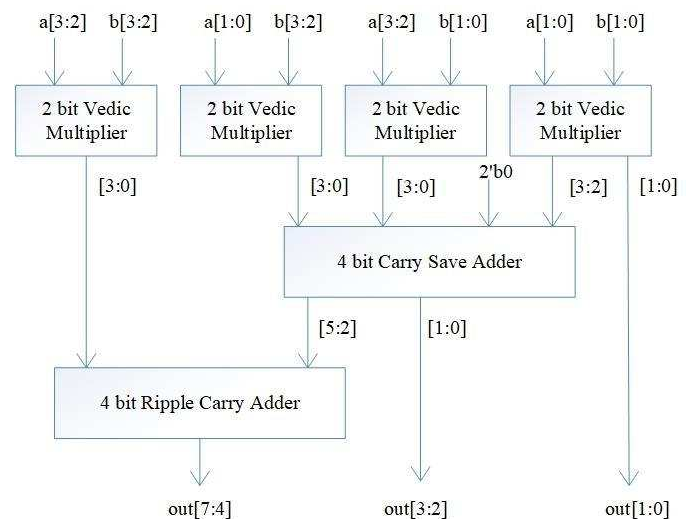


Fig. 4. Proposed 4 bit Vedic multiplier using 4 bit CSA and 4 bit ripple carry adder.

The 2 bit Vedic multiplier is implemented as shown in Fig. 5 where $a[1:0]$ and $b[1:0]$ are the 2 bit inputs and $out[3:0]$ is a 4 bit output respectively.

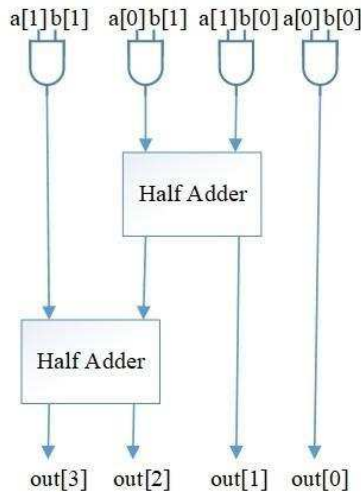


Fig. 5. 2 bit Vedic Multiplier

The Carry Save Adder can add three or more inputs and produces partial outputs in first step of addition. These partial outputs are added again and again in multiple stages to get the final sum where the number of steps depend on the inputs. A 4 bit CSA is shown in the Fig. 6.

Carry look ahead adder is prefixed in a parallel manner to get the Spanning tree adder which is shown in Fig. 7. Carry propagate bits and carry generate bits are computed only for three stages in the spanning tree adder. The carry propagate and carry generate bits are computed in the first stage using the input bits, (Fig. 8) and computed for succeeding stages from preceding stage outputs (Fig. 9). $pm[i]$ represents the carry propagate bit and $gm[i]$ represents the carry generate bit. AND, XOR, OR represent and, xor, or operations respectively.

The input bits are added by 4 bit carry select adders and the input carry to the least significant 4 bit carry select adder is the input carry given to the spanning tree adder and the input carry to the three most significant 4 bit carry select adders are generated from the carry propagate and carry generate bits of the third stage as per the computation in Fig. 10. So these three most significant carry select adders need not wait to perform addition on the input bits until the previous carry select adders complete the addition and generate the output carry.

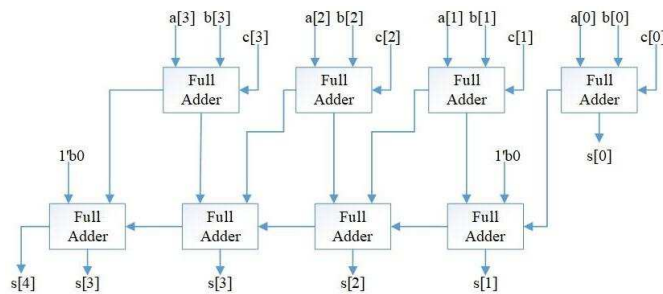


Fig. 6. 4 bit Carry Save Adder.

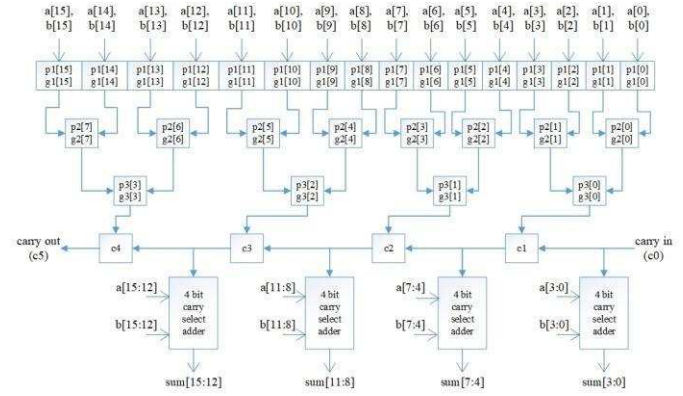


Fig. 7. 16 bit Spanning tree Adder.

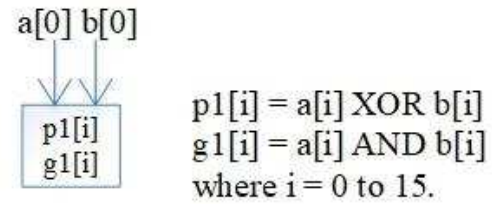


Fig. 8. Computation to produce carry propagate and carry generate bits of first stage of spanning tree adder.

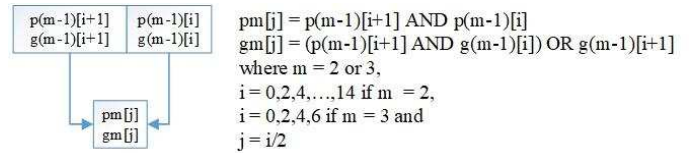


Fig. 9. Computation to produce carry propagate and carry generate bits of second and third stages of spanning tree adder.

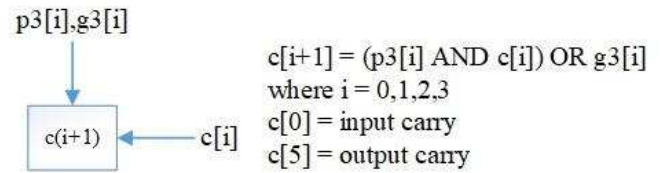


Fig. 10. Computation to produce input carry to the three most significant 4 bit carry select adders.

IV. RESULTS AND DISCUSSIONS

The proposed 16 bit MAC unit, 16 bit Vedic multiplier and existing 16 bit Vedic multiplier are simulated in ISIM simulator using Xilinx Vivado 2018.3 and the simulation outputs are shown in Fig. 11, 12 and 13 respectively. It is observed that, the propagation delay obtained for the multiplication in the proposed 16 bit MAC unit (18.762ns) is reduced compared to that of the existing 16 bit multiplier (25.684ns) because of the presence of spanning tree adder.

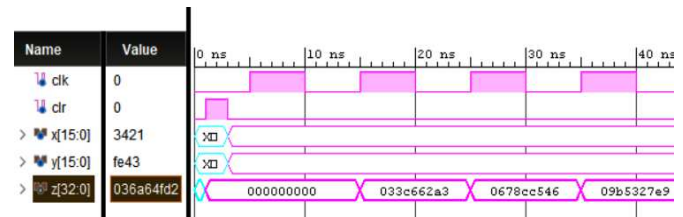


Fig. 11. Simulation output of the proposed 16 bit MAC unit in ISIM simulator.

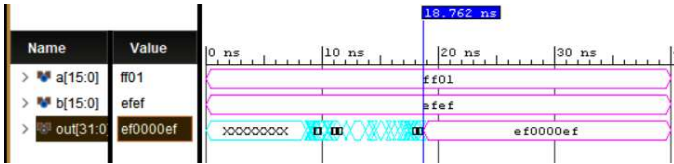


Fig. 12. Simulation output of the proposed 16 bit Vedic multiplier in ISIM simulator.

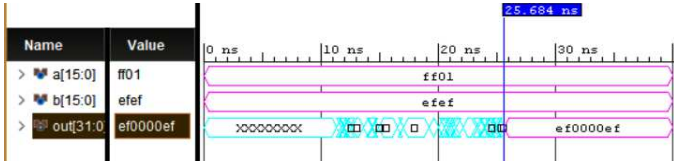


Fig. 13. Simulation output of the existing 16 bit Vedic multiplier in ISIM simulator.

The implementation results such as power, worst case combinational delay and the no. of LUTs utilized are observed in the Xilinx Vivado tool for both the existing and proposed designs for the Zedboard and are shown in the Table I. The observed power and delay in the proposed design are reduced to a good extent when compared to the existing design made up of CSAs.

The comparison of the proposed design with existing design in terms of power and delay are shown in Fig. 14 and 15 respectively.

TABLE I. POWER, DELAY AND AREA OF THE DESIGNS

Design	Power (W)	Delay (ns)	LUTs
Existing Design	0.489	30.894	474
Proposed Design	0.386	21.628	476

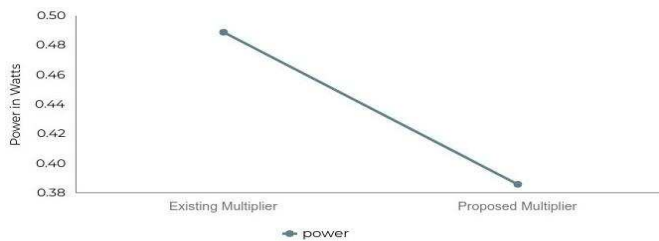


Fig. 14. Graph representing the variation in power (W) for existing and proposed designs.

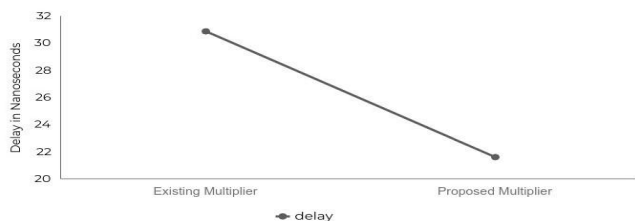


Fig. 15. Graph representing the variation in delay (ns) for existing and proposed designs.

V. CONCLUSION

The MAC unit plays a vital role in various signal processing computations. This paper presents a 16 bit MAC unit using 16 bit Vedic multiplier (UT algorithm) and a 32 bit

spanning tree adder. In this design, the 16 bit Vedic multiplier is built with 8 bit Vedic multipliers, 8 bit CSA along with 8 bit spanning tree adder. From the results of simulation and implementation, it is noticed that there is an improvement of 30% in delay and 21.1% in power compared with the existing 16 bit Vedic multiplier which used 8 bit Vedic multipliers and two 16 bit CSAs. The results show that the proposed MAC unit finds its application in high speed and power optimized signal processors. The proposed design can be further implemented using reversible logic gates, approximate computing to reduce the power, delay and area.

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